# LendSight

## LOAN INTELLIGENCE & CREDIT RISK ANALYTICS

POWER BI PROJECT DOCUMENTATION

An interactive Power BI report for loan application volume, funding, status comparisons and portfolio segmentation.

Technology: Microsoft Power BI Desktop | Excel | Power Query | DAX

**Dataset: synthetic practice data, not actual lending activity.**

| 22,253 | 23 | 2 | 2026-2027 |
|---|---|---|---|
| synthetic loan records | source columns | report pages | synthetic date window |

# 01 Project Overview

LendSight is an interactive Power BI report that consolidates loan application counts, principal amounts, supplied payment totals, interest rates, debt-to-income (DTI), loan status, and loan/borrower segments.

## Business context

The report brings key measures and categories together so users can review portfolio size, funding, the supplied payment field, status distribution, and composition by purpose, term, and home ownership.

## Project objectives

| Objective |
|---|
| Summarise application volume and principal represented by loan records. |
| Compare funded amount with the supplied payment field. |
| Show average interest rate and average DTI in the active filter context. |
| Compare loan-status groups using a documented project rule. |
| Explore issue-month activity, term, home ownership, purpose and state-code categories. |

## Report structure

| Page | Purpose | Main visuals |
|---|---|---|
| Summary | Portfolio snapshot | KPI cards; Good/Bad application split; funded vs. received comparisons; status metrics. |
| Overview | Trends and segmentation | Monthly applications; term; home ownership; loan purpose; state-code visual area. |

**Scope:** LendSight is descriptive analytics. It is not a credit scoring, loan approval, or automated lending decision system.

# 02 Dataset and Field Dictionary

Source workbook: **LendSight_Synthetic_Loan_Data_22253_Indian_States.xlsx**. Power BI table: **financial_loan**.

| Property | Detail |
|---|---|
| Data records | 22,253 (excluding header) |
| Source columns | 23 |
| Date fields | ISSUE_DATE, LAST_CREDIT_PULL_DATE, LAST_PAYMENT_DATE, NEXT_PAYMENT_DATE |
| Date window | 2026-2027; generated practice dates |
| Data status | Synthetic; not real customer or lender records |

## Field dictionary

| Field | Type | Meaning / use |
|---|---|---|
| ISSUE_DATE | Date | Issue/record date; main time-series field. |
| ID | Identifier | Synthetic row/loan ID; validate uniqueness before distinct counting. |
| PURPOSE | Category | Stated purpose, such as debt consolidation, credit card, car, house or medical. |
| VERIFICATION_STATUS | Category | Verified, Source Verified or Not Verified. |
| GRADE | Category | Loan grade A-G; used for grade segmentation. |
| HOME_OWNERSHIP | Category | Housing category such as MORTGAGE, RENT or OWN. |
| ADDRESS_STATE | Category | Two-letter state/UT-style code assigned synthetically. |
| LOAN_STATUS | Category | Fully Paid, Current, Charged Off, late states, grace period or Default. |
| EMP_LENGTH | Category | Reported employment length. |
| EMP_TITLE | Text/category | Employment title; potentially high-cardinality text. |
| LAST_CREDIT_PULL_DATE | Date | Synthetic latest credit-pull date. |
| LAST_PAYMENT_DATE | Date | Synthetic last-payment date. |
| NEXT_PAYMENT_DATE | Date | Synthetic next-payment date. |
| MEMBER_ID | Identifier | Synthetic member ID; validate uniqueness before member-level analysis. |
| SUB_GRADE | Category | Detailed grade, e.g. C5 or B4. |
| TERM | Category | Loan duration; observed values are 36 months and 60 months. |
| ANNUAL_INCOME | Numeric | Synthetic annual income; currency unit is not documented. |
| DTI | Ratio | Stored as decimal; 0.1656 represents 16.56%. |
| INSTALLMENT | Numeric | Scheduled instalment amount; unit is not documented. |
| INT_RATE | Ratio | Stored as decimal; 0.1320 represents 13.20%. |
| LOAN_AMOUNT | Numeric | Loan principal/funded amount represented by a record. |
| TOTAL_ACC | Count | Synthetic total-account count associated with the borrower record. |
| TOTAL_PAYMENT | Numeric | Simulated amount paid/received; not an audited repayment ledger. |

**Data note:** State/UT-style codes are randomly assigned, dates are generated for 2026-2027, payment totals are simulated, and the workbook does not specify currency. The dataset is suitable for visualisation practice, not real-world lending conclusions.

# 03 Data Model and Calculation Logic

## Preparation and model design

| Stage | Implementation |
|---|---|
| Source | Load the Excel workbook and select financial_loan. |
| Validation | Confirm 22,253 records and all 23 source columns after refresh. |
| Data types | Date fields as Date; amounts/ratios as numeric; identifiers and categories as appropriate text/category fields. |
| Date table | Create a continuous Calendar table and mark it as the date table. |
| Relationship | Active Calendar[Date] to financial_loan[ISSUE_DATE] relationship for time intelligence. |
| Identifiers | Keep ID and MEMBER_ID as identifiers; do not sum them. |

## Core measures

| Measure | DAX definition |
|---|---|
| Total Loan Applications | Total Loan Applications = COUNTROWS(financial_loan) |
| Total Funded Amount | Total Funded Amount = SUM(financial_loan[LOAN_AMOUNT]) |
| Total Amount Received | Total Amount Received = SUM(financial_loan[TOTAL_PAYMENT]) |
| Average Interest Rate | Average Interest Rate = AVERAGE(financial_loan[INT_RATE]) |
| Average DTI Ratio | Average DTI Ratio = AVERAGE(financial_loan[DTI]) |

## Status grouping

The project groups Fully Paid + Current as Good Loan Applications. Other statuses are grouped as Bad Loan Applications. This rule is specific to this demonstration.

```
Good Loan Applications = CALCULATE([Total Loan Applications], financial_loan[LOAN_STATUS] IN {"Fully Paid", "Current"})
```

```
Bad Loan Applications = CALCULATE([Total Loan Applications], NOT (financial_loan[LOAN_STATUS] IN {"Fully Paid", "Current"}))
```

```
Good Loan % = DIVIDE([Good Loan Applications], [Total Loan Applications])
Bad Loan % = DIVIDE([Bad Loan Applications], [Total Loan Applications])
```

## Date table and MTD

```
Calendar = CALENDAR(MIN(financial_loan[ISSUE_DATE]), MAX(financial_loan[ISSUE_DATE]))
```

```
Loan Applications MTD = TOTALMTD([Total Loan Applications], 'Calendar'[Date])
```

Format INT_RATE, DTI and the status percentages as percentages; do not multiply percentage results by 100 inside DAX.

# 04 Report Page: Summary

A compact view of the selected loan portfolio. KPI values respond to active grade and purpose filters.

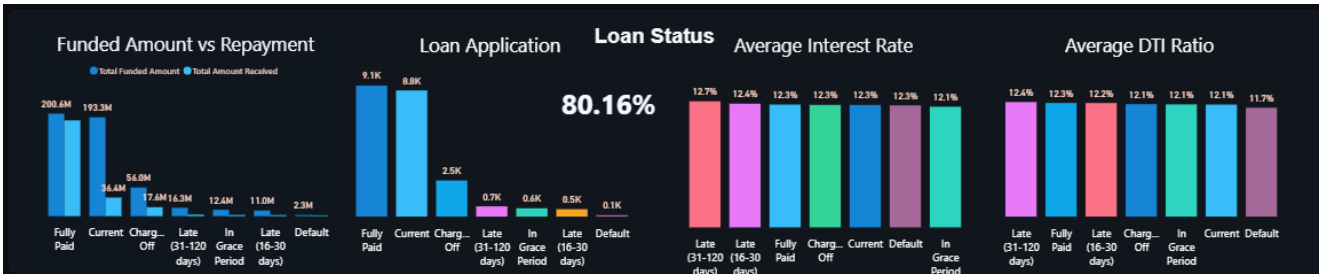
## Key portfolio indicators




*KPI strip from the supplied report screenshot.*

| KPI | Meaning | Reference calculation |
|---|---|---|
| Total Loan Applications | Records in current filter context. | COUNTROWS(financial_loan) |
| Total Funded Amount | Sum of loan principal. | SUM(financial_loan[LOAN_AMOUNT]) |
| Total Amount Received | Supplied payment field; simulated. | SUM(financial_loan[TOTAL_PAYMENT]) |
| Average Interest Rate | Average quoted rate. | AVERAGE(financial_loan[INT_RATE]) |
| Average DTI Ratio | Average debt-to-income ratio. | AVERAGE(financial_loan[DTI]) |

## Loan-status comparisons



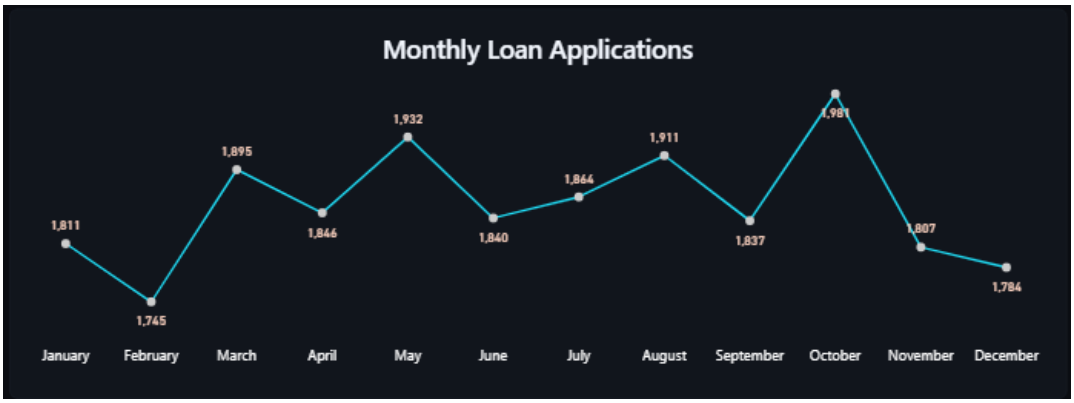

*Visual comparisons: funded amount versus supplied payment value, application counts, average interest rate and average DTI by status.*

In the documented dataset, Fully Paid + Current account for 17,839 records. Other statuses account for 4,414 records (19.84%).

# 05 Report Page: Overview

Explores issue-month activity and the distribution of records by term, home ownership, purpose and the synthetic state-code field.

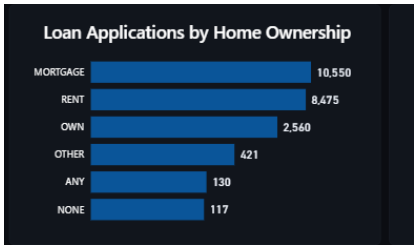
## Monthly loan applications



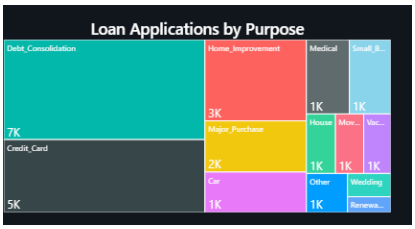

The monthly axis should preserve issue year so the same month in 2026 and 2027 is not unintentionally combined.




*Loan term*




*Home ownership*




*Loan purpose*

## Analytical views

| Visual | Question supported |
|---|---|
| Monthly applications | How does record volume vary over issue time? |
| Loan term | How are records distributed between 36- and 60-month terms? |
| Home ownership | Which housing categories are most represented? |
| Loan purpose | Which stated purposes account for the largest shares? |
| State-code distribution | How are synthetic state/UT-style labels distributed? |

ADDRESS_STATE values are synthetic labels and do not represent verified borrower geography.

# 06 Interaction and Visual Design

## Report interactions

| Feature | Behaviour |
|---|---|
| Grade slicer (A-G) | Filters records by GRADE. |
| Purpose slicer | Filters records by PURPOSE. |
| Summary / Overview navigation | Switches between report pages. |
| Reset Filters | Uses a saved bookmark intended to restore the default filter state. |
| Cross-filtering | Visual selections affect related visuals according to configured interactions. |

## Visual design system

| Design role | Colour | Use |
|---|---|---|
| Page background | #05070B | Dark canvas. |
| Visual cards | #101720 | Separate analytical panels. |
| Primary blue | #009DFF | Main bar and selected-state emphasis. |
| Bright cyan | #20D9F2 | KPI values and active highlights. |
| Positive accent | #00B7FF | Good/performing segment. |
| Risk accent | #FF4357 | Bad/risk group accent. |
| Borders / text | #16415C / #F4F7FB | Panel outlines and readable labels. |

## Project scope

LendSight demonstrates data preparation, DAX measures, date modelling, interactive filtering, KPI design and segmentation in Power BI. Synthetic records, generated dates, randomly assigned state-code categories and simulated payment totals mean the report demonstrates analytical techniques rather than real lending performance. The workbook does not define currency units.